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# General Knowledge
# World Geography

## STUDY NOTES

## Climatology: Koeppe N's Scheme Of Classification Of Climate

# Climatology: Koeppe N's Scheme Of Classification Of Climate

The world climate can be studied by organizing data on temperature, precipitation, and other climate-related factors. This helps in synthesizing information into smaller units for easier understanding, description, and analysis. There are three broad approaches to climate classification: empirical, genetic, and applied.

1. **Empirical Classification**: Based on observed data, particularly temperature and precipitation.
2. **Genetic Classification**: Organizes climates based on their causes.
3. **Applied Classification**: Designed for specific purposes.

## Koeppen's Scheme of Classification of Climate

The most widely used classification of climate is the empirical classification scheme developed by **V. Koeppen**. Koeppen identified a close relationship between the distribution of vegetation and climate. By selecting specific **values of temperature and precipitation**, he classified climates based on the distribution of vegetation. This scheme, developed in 1918 and modified over time, remains in use today.

**Koeppen** recognized **five major climatic groups**, with four based on temperature and one on precipitation. These groups are further subdivided into types based on the seasonality of precipitation and temperature characteristics.

| *Group* | *Characteristics* |
|---|---|
| A - Tropical | Average temperature of the coldest month is 18° C or higher |
| B - Dry Climates | Potential evaporation exceeds precipitation |
| C - Warm Temperate | The average temperature of the coldest month of the (Mid-latitude) climates years is higher than minus 3°C but below 18°C |
| D - Cold Snow Forest Climates | The average temperature of the coldest month is minus 3° C or below |
| E - Cold Climates | Average temperature for all months is below 10° C |
| H - High Land | Cold due to elevation |

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## Koeppen's Classification - Climatic Types

| Group | Type | Letter Code | Characteristics |
|---|---|---|---|
| A-Tropical Humid Climate | Tropical wet | Af | No dry season |
| | Tropical monsoon | Am | Monsoonal, short dry season |
| | Tropical wet and dry | Aw | Winter dry season |
| B-Dry Climate | Subtropical steppe | BSh | Low-latitude semi arid or dry |
| | Subtropical desert | BWh | Low-latitude arid or dry |
| | Mid-latitude steppe | BSk | Mid-latitude semi arid or dry |
| | Mid-latitude desert | BWk | Mid-latitude arid or dry |
| C-Warm temperate (Mid-latitude) Climates | Humid subtropical | Cfa | No dry season, warm summer |
| | Mediterranean | Cs | Dry hot summer |
| | Marine west coast | Cfb | No dry season, warm and cool summer |
| D-Cold Snow-forest Climates | Humid continental | Df | No dry season, severe winter |
| | Subarctic | Dw | Winter dry and very severe |
| E-Cold Climates | Tundra | ET | No true summer |
| | Polar ice cap | EF | Perennial ice |
| H-Highland | Highland | H | Highland with snow cover |

## Climatic Groups and Types

### Group A: Tropical Humid Climates

Tropical humid climates are located between the Tropic of Cancer and Tropic of Capricorn. These regions are characterized by the sun being overhead throughout the year, and the presence of the Inter Tropical Convergence Zone (ITCZ), which makes the climate hot and humid.

**Types of Tropical Humid Climates:**

1. **Tropical Wet Climate (Af)**
   - Found near the equator, with rainfall throughout the year.
   - Temperatures remain uniformly high.
   - Example: Amazon Basin, western equatorial Africa.
   - Vegetation: Tropical evergreen forests with dense canopy cover and high biodiversity.
2. **Tropical Monsoon Climate (Am)**
   - Found in parts of India, northern Australia, and South America.

- Heavy rainfall occurs mostly during the summer.
- Winter is dry.
- Example: Indian Subcontinent.

3. **Tropical Wet and Dry Climate (Aw)**
   - Occurs north and south of the tropical wet climates.
   - Borders with dry climates on the west and humid climates on the east.
   - Example: Brazil, Sudan, South Africa.

## Group B: Dry Climates

Dry climates are characterized by very low rainfall, which is insufficient for the growth of plants. These climates are common in large areas, especially between latitudes 15° and 60° north and south of the equator.

**Types of Dry Climates:**

1. **Subtropical Steppe (BSh)**
   - The transition zone between humid and dry climates.
   - Receives slightly more rainfall than deserts, supporting sparse grasslands.
2. **Subtropical Desert (BWh)**
   - Hot, arid regions with very little rainfall.
   - Example: Sahara Desert, Arabian Peninsula.
3. **Mid-latitude Steppe (BSk)**
   - Semi-arid climate found in the interiors of continents.
   - Examples: Central Asia, and parts of North America.
4. **Mid-latitude Desert (BWk)**
   - Very dry, with harsh winters.
   - Example: The Great Basin in North America.

## Group C: Warm Temperate (Mid-latitude) Climates

These climates are found between latitudes 30° and 50° and are characterized by warm summers and mild winters.

**Types of Warm Temperate Climates:**

1. **Humid Subtropical Climate (Cwa)**

- Occurs on the poleward side of tropical regions.
- Hot, humid summers with dry winters.
- Example: Eastern China, North Indian plains.

2. **Mediterranean Climate (Cs)**
   - Hot, dry summers and mild, rainy winters.
   - Example: California, parts of Chile, and South Australia.
3. **Humid Subtropical (Cfa) Climate**
   - Found in areas with year-round rainfall.
   - Example: Eastern United States, southern China.
4. **Marine West Coast Climate (Cfb)**
   - Located on the west coast of continents, characterized by mild temperatures and rainfall throughout the year.
   - Example: Northwestern Europe, Pacific Northwest of North America.

## Group D: Cold Snow Forest Climates

Cold snow forest climates occur between latitudes 40° and 70° in the northern hemisphere, especially in regions of North America, Europe, and Asia.

**Types of Cold Snow Forest Climates:**

1. **Cold Climate with Humid Winters (Df)**
   - Long, cold winters with heavy snow.
   - Example: Parts of Canada, Northern Russia.
2. **Cold Climate with Dry Winters (Dw)**
   - Very cold winters with low precipitation.
   - Example: Northeastern Asia.

## Group E: Polar Climates

Polar climates are found beyond 70° latitude, where temperatures are consistently low throughout the year.

**Types of Polar Climates:**

1. **Tundra Climate (ET)**
   - Characterized by permafrost, low-growing vegetation, and long summer days.

- Example: Arctic regions.

2. **Ice Cap Climate (EF)**
   - Found in regions like Greenland and Antarctica, with temperatures below freezing even in summer.
   - Very little precipitation.

### Group H: Highland Climates

Highland climates are influenced by the elevation and topography of mountainous regions. They are marked by large temperature variations over short distances and significant differences in precipitation based on elevation.

## Conclusion

Understanding Koeppen's classification helps in recognizing the relationship between climate and vegetation, providing insights into the distribution of ecosystems across the planet. By dividing climates into broad groups and types, Koeppen's system makes it easier to analyze the complexities of global climate patterns.

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